

Anany Sharma

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EDUCATION

University of Florida

Masters of Science in Artificial Intelligence Systems | 3.9 GPA

Graduated: May 2026

SRM Institute of Science and Technology

Bachelors of Technology in Computer Science | 3.7 GPA

Graduated: July 2021

SKILLS

Languages Python · TypeScript/Javascript · C++ · Dart · Java · SQL · NoSQL

AI/ML PyTorch · TensorFlow · scikit-learn · OpenCV · Generative AI · NLP · Computer Vision · Claude · Codex · Cursor

MLOps & Data MLflow · Spark · Airflow · Kafka · PowerBI · Pinecone · ETL

Software & Cloud AWS · AgentCore · Azure AI Search · Terraform · Docker · Kubernetes · REST · React.Js · Jenkins · Git

EXPERIENCE

AI Research Fellow (NSF Funded)

UNIVERSITY OF FLORIDA

March 2025 - Present

Gainesville, Florida

- Engineered an **SLM** on **Jetson Orin**, with a model footprint of **96MBs**, exported using **ONNX** runtime and **TensorRT**
- Optimized training/inference by KV Cache, mixed precision training, flash attention achieving **70T/s** with **2.44ms** TTFT
- Engineered a volatility-aware **TCN + attention** model to hyper-locally forecast PM2.5 air quality across 5+ years of data
- Built an Edge ML pipeline on ESP32-S3 via a compressed DL model(Pytorch to C byte array) with a Huber loss of 0.26.

AI Engineer Intern

OELRICH CONSTRUCTION

Aug 2025 - April 2026

Gainesville, Florida

- Built a RAG system on Azure, with **hybrid search** (semantic+BM25), reranking, query expansion for retrieval accuracy
- Created custom ingestion pipelines to sync **Azure VM**(F: Drive) data into embedded vectorized records into **CosmosDB**
- Designed an LLM evaluation framework using **DeepEval**, integrating unit tests, LLM-as-a-Judge scoring and A/B testing
- Achieved **.74 Recall@Top10**, **.78 NDCG@Top10** with an **F1 Score** of **.82** across all failure mode in DeepEval evaluation

Software Engineer - Data/AI

UNITED HEALTH GROUP

Jan 2022 - Aug 2024

Noida, India

- Engineered a **RAG** system with **Azure Blob Storage** as knowledge base and **Azure AI Search** for indexing and retrieval
- Improved retrieval **Recall@10** by 12% via semantic chunking for logical data partitioning, query rewriting and re-ranking
- Designed **Power BI** business KPI dashboards using **Oracle SQL** in DirectQuery with push-based data refresh
- Automated **1.5K+** biweekly financial reports using **SSIS/SSRS** with data validation and reconciliation workflows
- Architected end to end ETL pipelines using IBM WTX, Oracle SQL to process **10M+** monthly claims on UNIX AIX servers

Machine Learning Engineering Intern

MFIT TECHNOLOGIES

July 2020 - Nov 2020

- Developed an OCR/NLP enabled financial data system with **85% field-extraction accuracy** for transaction monitoring
- Engineered document extraction system supporting **10+ formats** using hybrid NER and layout-aware frameworks
- Built dynamic **Conditional Random Field** plus **Spatial** model to parse financial statements across **4 major banks**

PROJECTS

SMIRE AI - Medical Multi AI Agent System | Individual Project - [GITHUB](#)

Feb 2025 - May 2025

- Architected a **RAG service** using **Pinecone** to enable conversational access to medical reports and prescriptions
- Engineered an agentic system with **CrewAI** and **LangChain**, leveraging **Chain of Thought (CoT)** and **ReAct** prompts
- Integrated an appointment booking system with **Twilio** and a **calendar API** to automate personalized scheduling
- Deployed the full-stack **React/Flask** platform via Docker on Vercel/Railway, with **Prometheus** for system monitoring

EMOGEN - Emotion Aware Multilingual Speech to Image Generation | Individual Project - [GITHUB](#)

Mar 2025 - April 2025

- Engineered a multi-modal generative AI pipeline to integrate speech, emotion for **contextual image synthesis**
- Fine-tuned a **Whisper-based ASR** model, achieving **~50% WER** reduction using only 8 hours of low-resource language audio
- FineTuned a **CLIP-guided Stable Diffusion** model for emotionally (**HumeAI API**) and semantically aligned image generation

TRADE-MCP - Remote MCP Server for Real time Zerodha Trading | Individual Project - [GITHUB](#)

June 2024 - July 2024

- Developed and deployed a custom remote **Model Context Protocol (MCP)** server enabling GitHub Copilot and Cursor
- Integrated MCP server with **Zerodha Kite API** for real-time market data analysis, order placement, and portfolio insights
- Built backend using **Next.js** and **TypeScript**, deployed on Vercel for scalable, low-latency performance

PATENT

Crowd Detection and a Method Thereof - IIP(2021) | Published Patent (~80 hours) - [PATENT](#)

Jan 2021 - July 2021

- Developed and patented a **YOLOv5**-based multimodal crowd detection system using **Python**, **Flask**, and **OpenCV**, designed to identify high-risk crowd behavior and Mask/PPE compliance in real time
- Achieved up to **85% AP** for mask detection and **80%** for no-mask cases, optimized model performance with **transfer learning** and **mAP** evaluation, and deployed the solution with **Docker** for scalability